

SAMIA RAHMAN

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EDUCATION

University of Alberta

BSc Specialization in Computing Science

Edmonton, Alberta

Expected Graduation: May 2027

Relevant Coursework: Software Development and Product Management, Software Testing, Artificial Intelligence, Machine Learning

EXPERIENCE

Undergraduate Teaching Assistant

University of Alberta

Jan 2026 – Present

Edmonton, AB

- Facilitated Python-based lab sessions focused on automation, testing, and responsible AI usage, guiding students in building structured, reliable systems while emphasizing ethical and trustworthy application of AI tools.

Data Engineer Intern

City of Edmonton – Financial & Corporate Services

May 2025 – December 2025

Edmonton, AB

- Designed and deployed AI-driven automation workflows integrating 20 SAP Ariba REST APIs using Python (Pandas, NumPy, SQLAlchemy), SQL Server, and Gemini API to streamline supply chain reporting and administrative processes, transforming manual workflows into scalable systems supporting 500+ corporate users.
- Engineered reporting and analytics infrastructure with monitoring and data quality controls, enabling performance trend analysis and executive-level decision support while reducing reporting discrepancies by 30%.

Machine Learning Developer Co-op

Process Data Analytics and Smart Automation Lab (PDASA)

Jan 2025 – Apr 2025

Edmonton, AB

- Designed and deployed a real-time AI control system in PyTorch to monitor bitumen extraction processes and detect debris within Primary Separation Vessels (PSVs), integrating closed-loop valve regulation in collaboration with oil & gas industry partners to replace manual monitoring workflows.
- Implemented an A3C actor-critic architecture with CNN-LSTM encoders and structured performance monitoring pipelines, benchmarking against baseline PID control to improve response time, enhance operational stability, and reduce training time by 35% after migrating from TensorFlow.

Undergraduate Student Researcher – ML/Robotics

University of Alberta

Sep 2024 – Present

Edmonton, AB

- Designed a variational autoencoder (VAE) integrated with Soft Actor-Critic (SAC) in PyTorch to improve policy generalization and cross-task transfer in robotic control environments (Robosuite), reducing required training data by 40%.
- Optimized linear Model Predictive Control (MPC) for UAV navigation (thrust, pitch, roll) by refining feedback dynamics and controller tuning, achieving 27% faster response, 41% improved stability, and 18% lower computational overhead.

PROJECTS

UNTAPPED: Artist Management Web App | React, Node.js, PostgreSQL, OAuth2, AWS S3

Jan 2026 – Present

- Developing a full-stack web platform to manage artist onboarding, invoice workflows, payment tracking, and document storage using React front-end interfaces and RESTful Node.js APIs with PostgreSQL.
- Implementing role-based authentication, invoice approval state management, and secure AWS S3 file storage within Agile/Scrum sprints, replacing manual spreadsheet processes with a scalable SaaS-style architecture.

IoT Smart Home Test Automation | Java, JUnit 5, Docker, GitHub Actions

Dec 2025

- Designed TDD test suite using black-box and boundary value analysis, achieving 80%+ branch coverage for state consistency logic (heating/cooling exclusivity, sensor accuracy, alarm triggers).
- Built CI/CD pipeline with Docker and GitHub Actions to automate test execution and coverage analysis on every commit.

FoodMinder: Smart Grocery Management | TypeScript, React Native, Gemini API, Firebase

Feb 2025

- Built an LLM-powered React Native app using Google Gemini for receipt OCR and structured data extraction, reducing manual entry by 90% with real-time sync via Firebase.

TECHNICAL SKILLS

Full-Stack Development: Python, Java, C, TypeScript, JavaScript, SQL, R, React, React Native, Node.js, Express, HTML/CSS, REST APIs, PostgreSQL, MongoDB

Data Engineering & Analytics: ETL Pipelines, Pandas, NumPy, Power BI, Tableau, Jupyter Notebook

AI/ML: Machine Learning, Reinforcement Learning, Large Language Models (LLMs), Generative AI, Prompt Engineering, RAG, Embeddings, PyTorch, TensorFlow, Scikit-learn

Cloud & DevOps: AWS, Docker, Firebase, Linux, Git, GitHub Actions, CI/CD

EXTRACURRICULAR

VP Finance – Conferences, Ada's Team

May 2025 – Present

- Led conference finance and logistics as VP Finance for Ada's Team, managing \$45K+ in budgets to sponsor 40+ students, coordinating travel, sponsorships, and events for 200+ underrepresented CS participants.